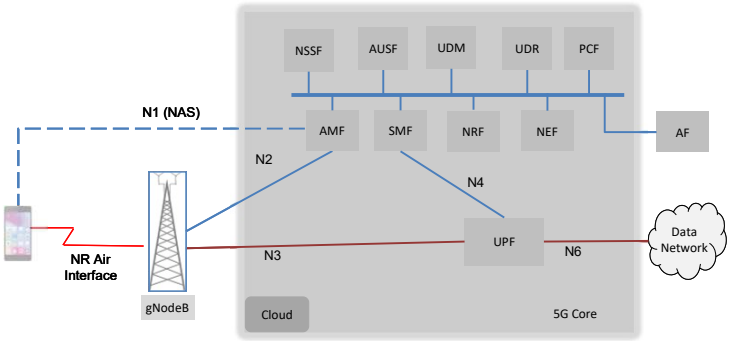
5G Network Overview



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Course Topics

- Overview of Mobile Networks and Evolution (2G to 5G Evolution)
- Overview of Wireless Technologies for Mobile Communication
- Overview of Network Architecture Evolution
- 5G Overview and 5G Advancements
- 5G Implementation Approaches – Overview
- 6G Overview - Expectations and Requirements

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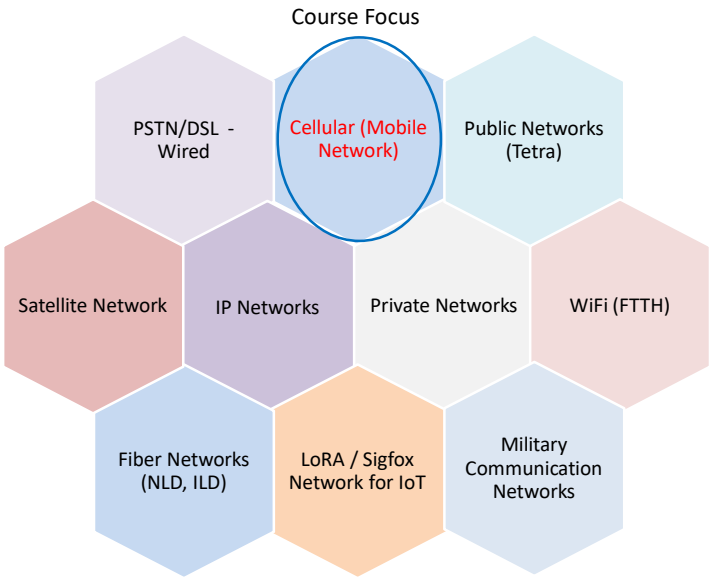
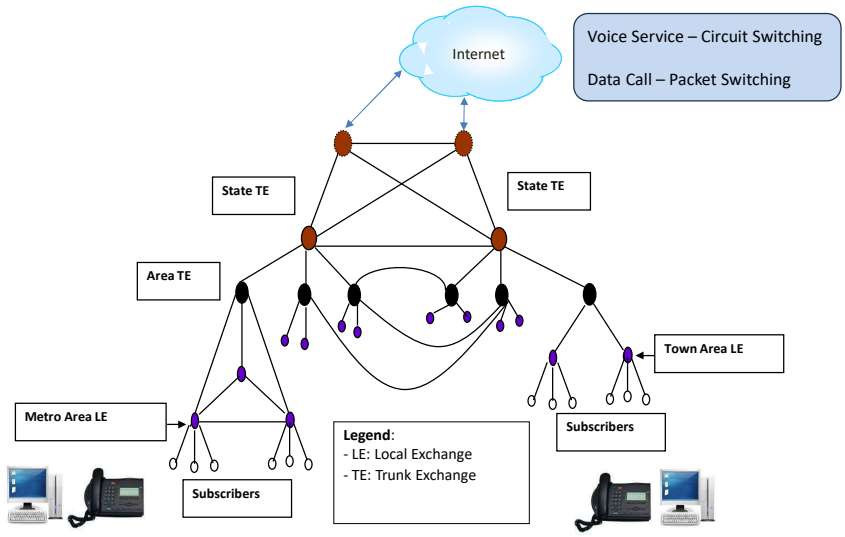
Mobile Network Evolution (2G to 5G)

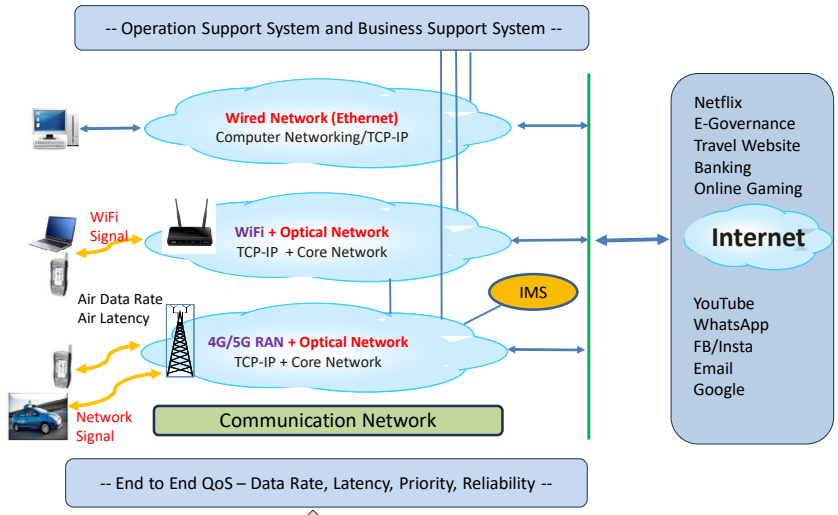
Communication Networks



PSTN / Exchange Hierarchy

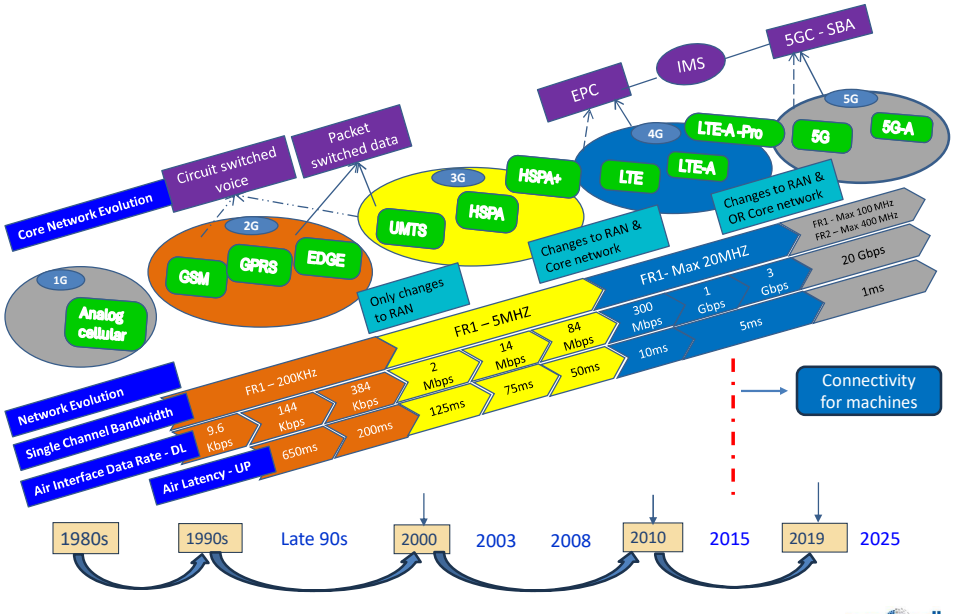


Various Options to Access Internet

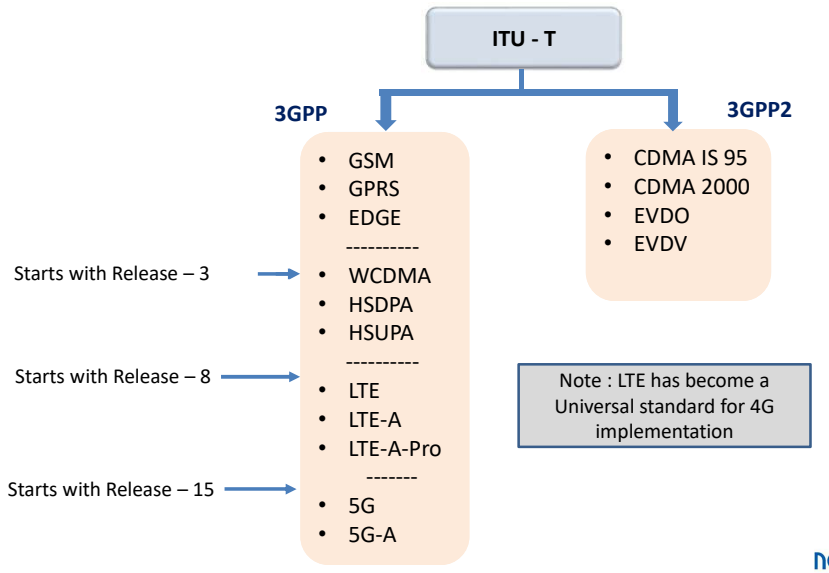


QoS Fundamentals: These parameters will change depending on type of application

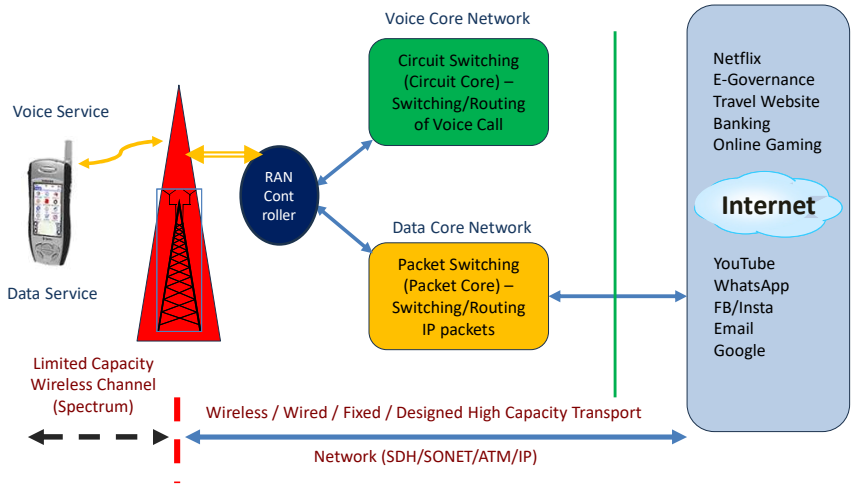
1G to 5G - 3GPP Evolution Path



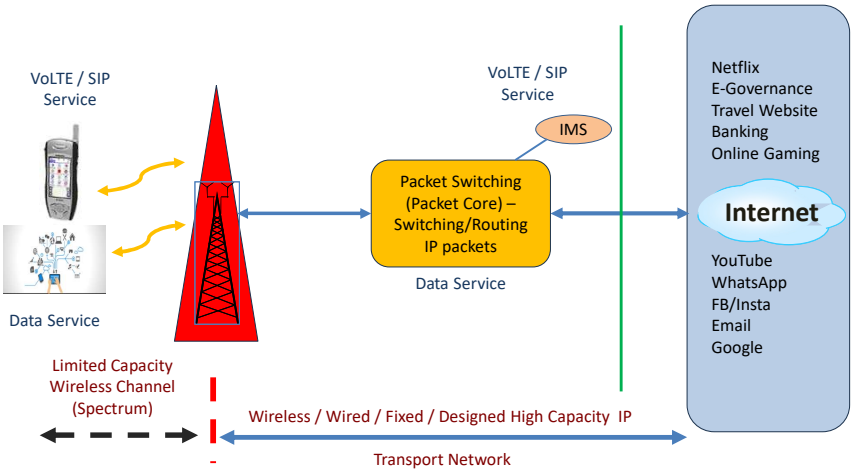
Technology Under ITU-T



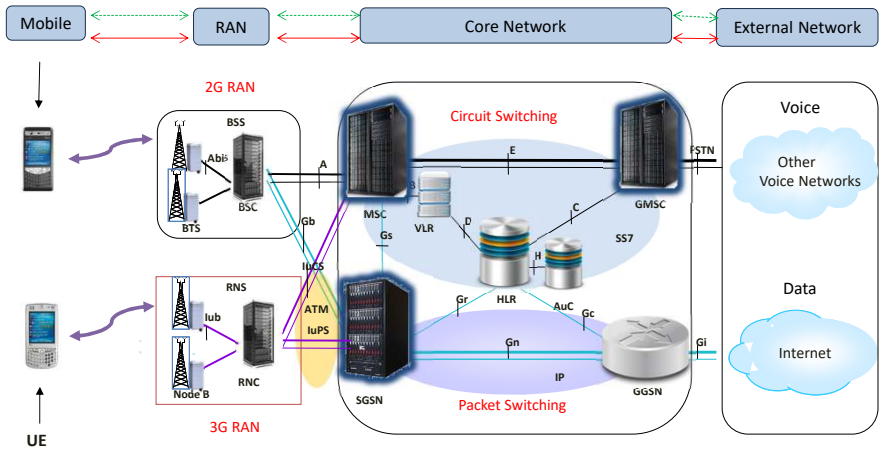
2G-3G Mobile Network – Big Picture



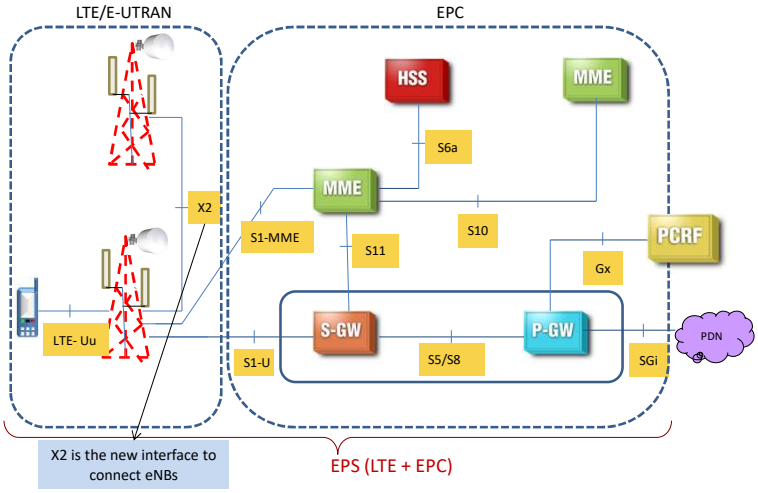
4G/5G Mobile Network – Big Picture



Mobile Network Architecture (ex- 2G and 3G)



4G (LTE-EPC) Network Architecture



MME	Mobility Management Entity	**HSS**	Home Subscriber Server	**PCRF**	Policy and Charging Rules Function	**PDN**	Packet Data Network
S-GW	Serving Gateway	**P-GW**	Packet Gateway	**eNodeB**	Evolved NodeB	**EPC**	Evolved Packet Core

Voice in 4G (LTE-EPC) Network

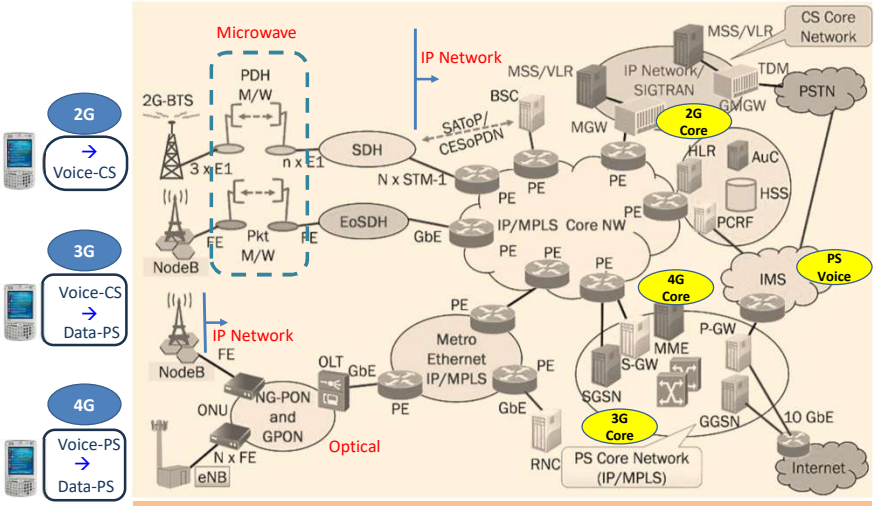
CSFB
(Circuit Switched Fall Back)

- Offload voice to 2G/3G networks.
- Quick, temporary fix.
- Used in current LTE deployment by 3GPP operators.
- Only CS voice is used.

VoLTE
(Voice Over LTE)

- Long term solution for voice.
- Use of IMS (IP Multimedia System)
- Voice in LTE carried as VOIP
- UE should support VoLTE
- Use of SIP Protocol

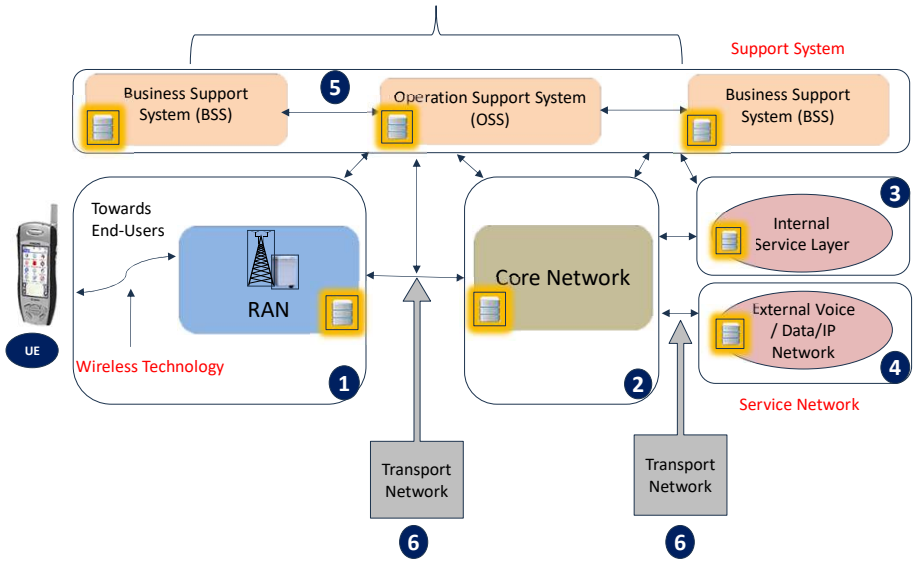
Mobile Networks – E-2-E View of Network



A end to end view of Transport Network (Practically 1000s of KM distance)

Picture Source: LinkedIn

Commercial Mobile Network – Big Picture



Sub Systems of mobile networks

Sub System	Key Points	Key Technologies
Radio Access Network (RAN)	*Channel Bandwidth, Radio Resource Management, Signalling and Data Bearer, Admission Control, Power Control, Paging, Multiple Access schemes, Antenna, Handover, Interference Management,*	GSM, GPRS, WCDMA, LTE, 5G NR wireless technologies
Switching Network (Core Network)	*Registration, Authentication, Connection Management, Session Management, Routing, Switching, Service Management, Mobility Management, QoS, Policies, User Data Base, Profiles, etc*	Circuit Switching, Packet Switching, IP Network
Transport Network (Backhaul, Midhaul)	*Interface between Network Components, transfer of voice/data traffic and signalling, Microwave, Fiber, Copper, Ethernet, Gigabit Ethernet, etc*	IP, SDH, PDH, DWDM, SONET, ATM, MPLS, GPON

Sub Systems of mobile networks

Sub System	Key Points	Key Technologies
Operations Support System (OSS) (NMS, EMS, NOC, OMC	*Activation, Deactivation, Fault, Performance, Configuration, Alarm, Security Management, Inventory Management, Network Optimization, Network O&M*	Java, Database, API Software Programming, Linux, Cloud, Diameter
Business Support System (BSS)	*CRM, Billing, Fraud Management, Revenue Management, Revenue Leakage Management, Interconnect Billing, Revenue Assurance, Intelligent Network, Subscriber Profiling, New Tariff Plans*	Java, Database, API Software Programming, Linux, Cloud, Diameter
Service Layer	*Applications for users, Voice Service, Operator Specific Services, SMS, MMS, Video, Internet Applications, etc*	SIP, IP, TCP/IP, App Development, Software Programming

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Network Operation Centre – A part of OSS

OMC-R:- Operation & Maintenance Center for Radio Network

OMC-S:- Operation & Maintenance Center for Switching Network

NOC:- Overall Network Operation Center (Regional or Global) which includes Radio, Switch , IN, VAS and Packet Core

FCAPS

- F:- Fault Management
- C:- Configuration Management
- A:- Accounting Management
- P:- Performance Management
- S:- Security Management

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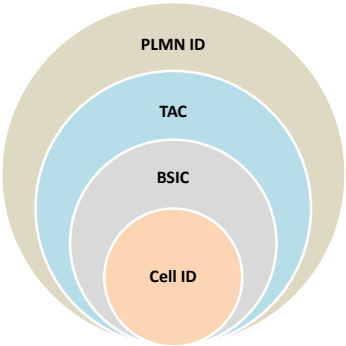
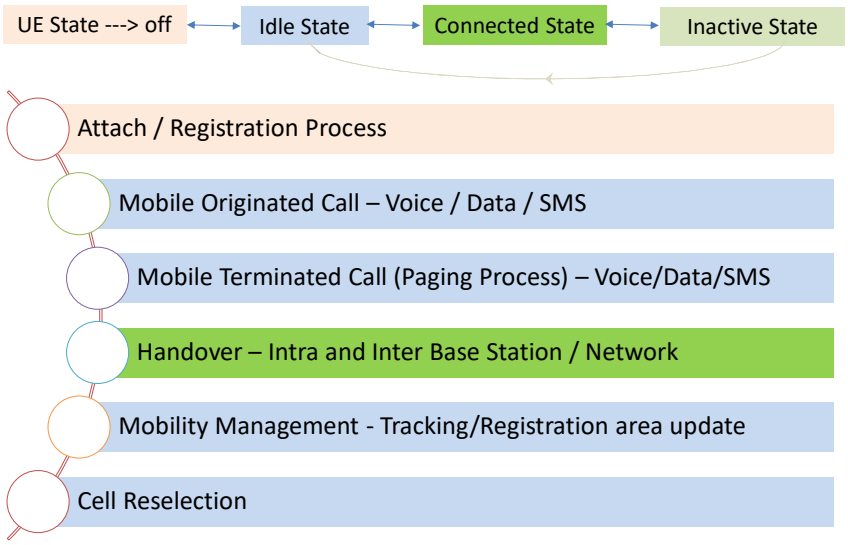
Cellular Network Identities



UE Identities in Mobile Network

- MSISDN (Identity of user – Voice Call)
- IMEI - (Handset)
- IMSI / S-TMSI (SIM)
- IP Address (Network Provided)
- RNTI (Radio Network Temporary Identifier) - Network Provided
- GUTI – Globally Unique Temporary Identifier

UE Process in Mobile Networks



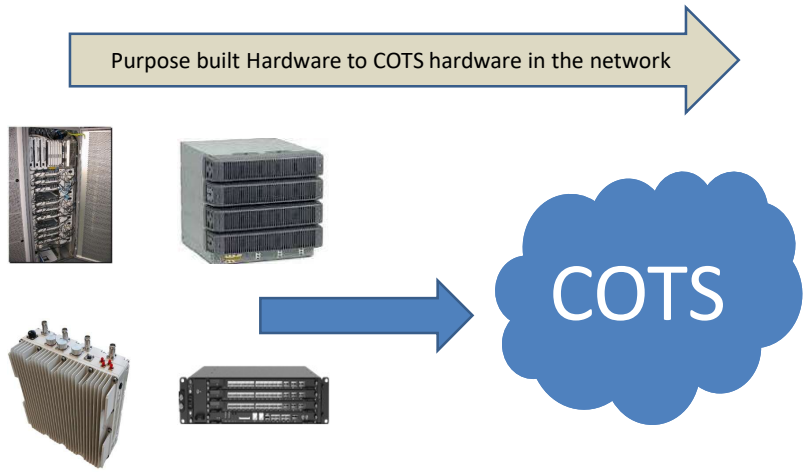
Telecom Industry Eco System

- Telecom Operators or Service Providers (E.g: Airtel, Vodafone, AT&T, Telefonica, BT, etc)
- Telecom Network OEMs or Vendor (E.g: Ericsson, Nokia, Ciena, Cisco, Huawei, etc)
- Telecom Regulatory Bodies (E.g: TRAI, FCC)
- Telecom Standardization Organizations (E.g: ITU, IEEE)
- Software Service Providers and Integrators (E.g: Wipro, TCS, Infosys, Accenture, THIS, etc)
- Subscribers and Corporate Customers
- Chipset Vendors (E.g: Qualcomm, Texas Instruments)
- Handset Vendors (E.g: Apple, Nokia)
- Mobile OS (E.g: Android, iOS)
- OSS and BSS Solution Providers (E.g: Amdocs, RCP, ATOS, etc)
- Infrastructure Providers (E.g: GCP, AWS, Azure, HP, IBM, etc)

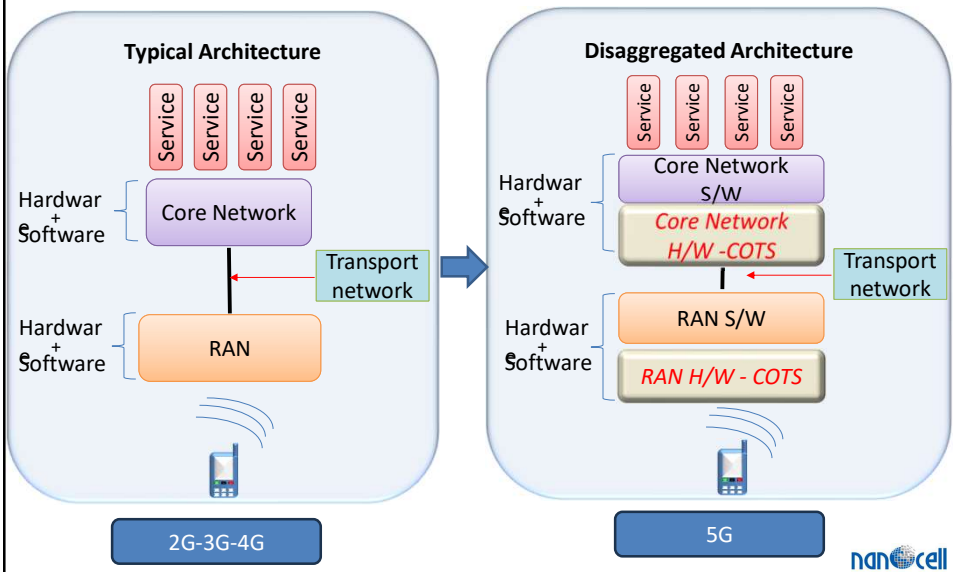
Evolution of Mobile Network Solution

Generation / Type of Network	Radio Network	Core Network	Network Interfaces	Dependency on OEM for Network Expansion/Upgrade
2G	HW+SW Based	HW+SW Based	Closed + Open Interfaces	High
3G	HW+SW Based	HW+SW Based	Closed + Open Interfaces	High
4G	HW+SW Based	SW Based + Specific Cloud (VNF)	Closed + Open Interfaces	Medium
5G	HW+SW Based – Cloud (CNF)	SW Based + Cloud (CNF)	Open Interfaces	Low
Network Slice	Virtual RAN	Virtual Core	Virtual Network	Low
5G NAAS				
5G Private				

What Telcos want

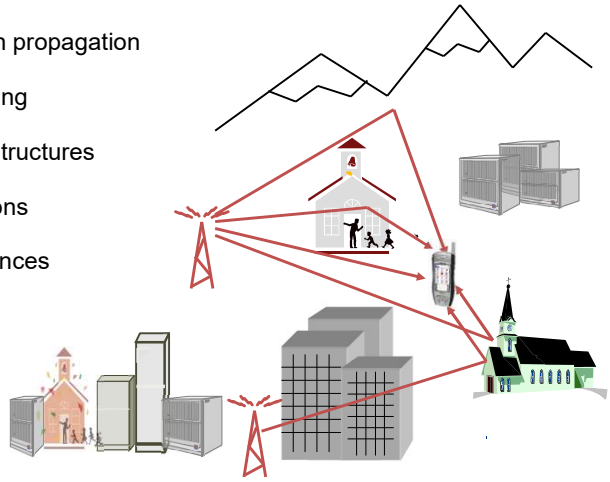


Mobile Network Solution – Evolution to 5G

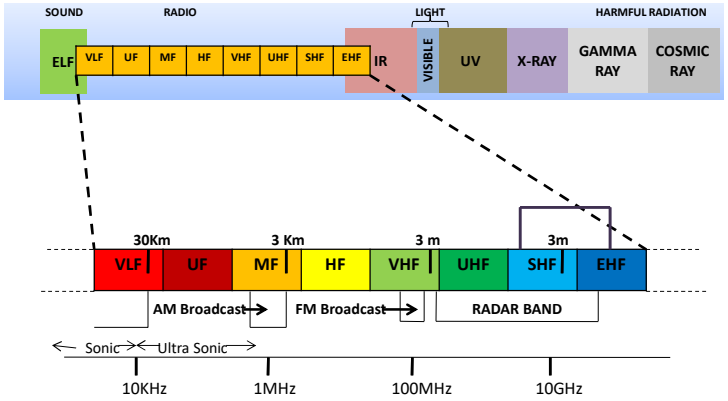


Radio/Wireless Environment

- Multipath propagation
- Shadowing
- Terrain structures
- Reflections
- Interferences



Electromagnetic Spectrum



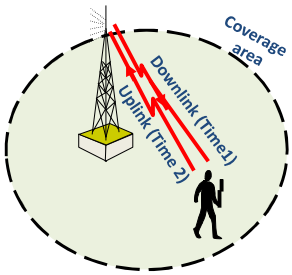
Spectrum and Multiple Access in Mobile Networks

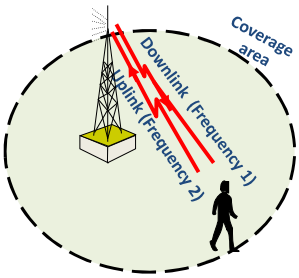
Wireless Technology	Frequency Band	Single Carrier Bandwidth	Multiple Access Scheme
GSM, GPRS, EDGE	900, 1800, 1900 MHz	200 KHz	TDMA
CDMA IS-95, 2000	800, 2100 MHz	1.25 MHz	CDMA
WCDMA, HSPA	2100MHz	5 MHz	WCDMA
LTE	Multiple Bands (Band 1 to 40)	1.4/3/5/10/15/20 MHz	OFDMA & SC-FDMA
5G (FR1)	450MHz – 7.125GHz	5 - 100 MHz	OFDMA & SC-FDMA (OFMDA)
5G (FR2-1) - mmWave	24.25GHz – 52.6GHz	50 - 400 MHz	OFDMA & SC-FDMA (OFDMA)

Duplexing – TDD/FDD

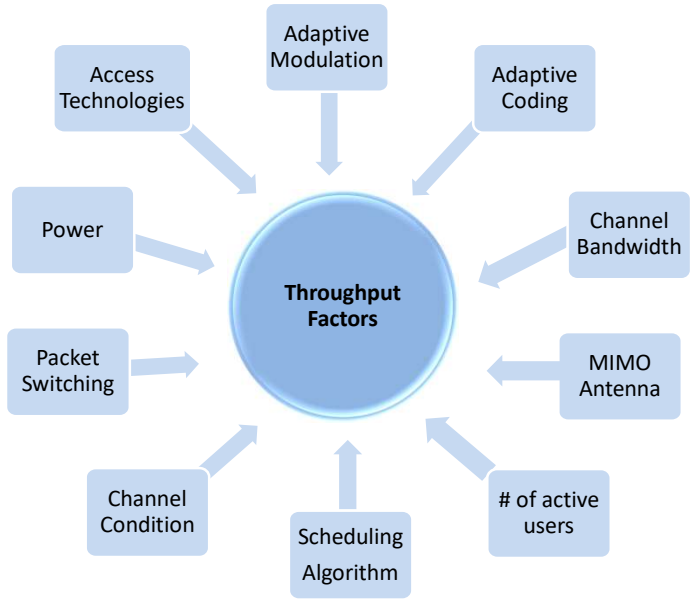
In Duplex communication system Uplink and Downlink are separated by two parameters

Time	Frequency





Factors That affect data rate in Wireless Channel



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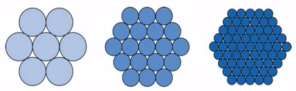
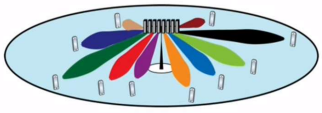
How do we get 1000 times more data rate

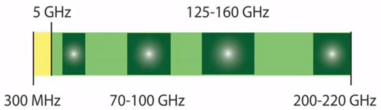
10x? • Higher cell density
– More access points per km²



20x? • Higher spectral efficiency
– Direct signals towards many users



5x? • More frequency spectrum
– More transmission per second
– Use much higher frequencies



Source: Communication Systems, Linköping University, LIU

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5G Overview and Motivation

LTE Evolution to 5G

LTE (Rel 8, 9)	• Rel-8 standard was frozen in March 2009. • Goal to prepare the mobile system to allow evolutionary change towards 4G as imposed by IMT-Advanced. • Pushed to market to counter WiMAX • Mobile Broadband was the focus
LTE-Advanced (Rel 10,11,12)	• The corresponding standard was frozen in June 2011. • LTE-Advanced was defined to fulfill IMT-Advanced requirements, thus is seen as a 4G technology. • Carrier Aggregation, Advanced MIMO, HetNet features were introduced
LTE-Advanced Pro (Rel 13, 14)	• This was agreed by 3GPP in October 2015 as a marker for LTE for Rel-13 onwards. • The new name is used to mark a point where significant improvements with regards to LTE-Advanced are made.

LTE-A Pro – Key Features (Step towards 5G)

3GPP Rel-13 and Rel-14

- Support for IoT Connectivity (eMTC, NB-IoT)
- Use of Unlicensed Band (LAA, LTE-WiFi Interworking)
- Better Network Capacity (massive MIMO Antenna, Carrier Aggregation - CA)
- Cell Edge Performance and Reliability (Dual Connectivity - DC)

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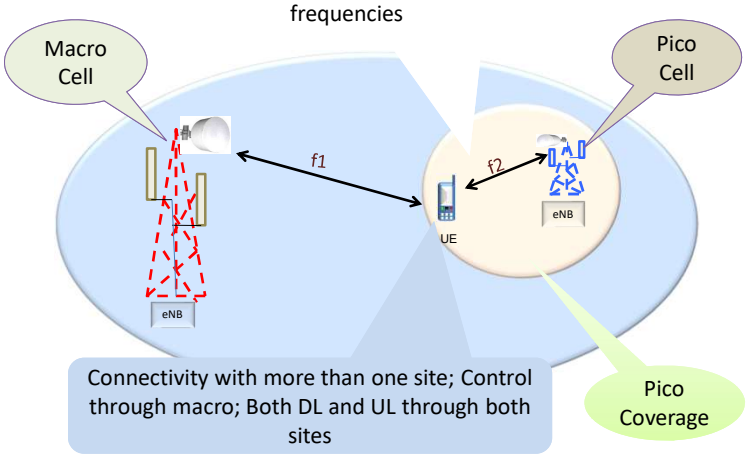
Dual Connectivity

Inter site Aggregation; multiple carrier frequencies



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IOT Connectivity on LTE Network – Evolution

3GPP Release	Release 8	Release 8	Release 12	Release 13	Release 13
4G UE Category	Cat4	Cat 1	Cat.0	Cat M1	Cat NB1
Downlink peak rate	150 Mbps	10 Mbps	1 Mbps	1 Mbps	200 Kbps
Uplink peak rate	50 Mbps	5 Mbps	1 Mbps	1 Mbps	144 Kbps
Number of antennas	2	2	1	1	1
Duplex mode	Full duplex	Full duplex	Half duplex	Half duplex	Half Duplex
UE receive bandwidth	20 MHz	20 MHz	20 MHz	1.4 MHz	200 KHz
UE transmit power	23 dBm	23 dBm	23 dBm	20 dBm	23 dBm
Modem Complexity	100%	80%	40%	20%	<15 %
				eMTC	NB-IOT

Scaling up in performance and mobility to suit broadband requirement

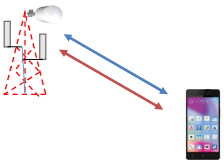
Scaling down in complexity and power to suit IoT connectivity requirement

Study Source : Online

Why 5G

LTE and its evolutions might not meet the broadband requirements post 2020

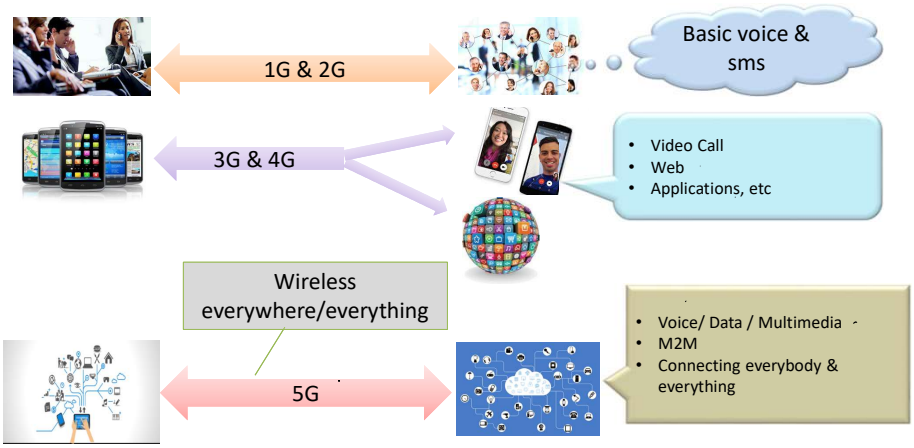
Smartphone type user saturation and falling ARPUs



How to get new businesses to use wireless and contribute to revenue and growth?.

Are there new areas? what are those? how is technology different? business models (?)

5G Objectives



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5G Requirements

Use Case Perspective

Security	Reliability	Latency
Experienced throughput	Availability	Traffic volume
Connection density	Cost	Energy Consumption
Peak rates	Mobility	

Network Perspective

- *Open Network*
- *Flexible Capacity*
- *Software Based*
- *Highly Automated*
- *Self Organized*
- *Easy to upgrade*

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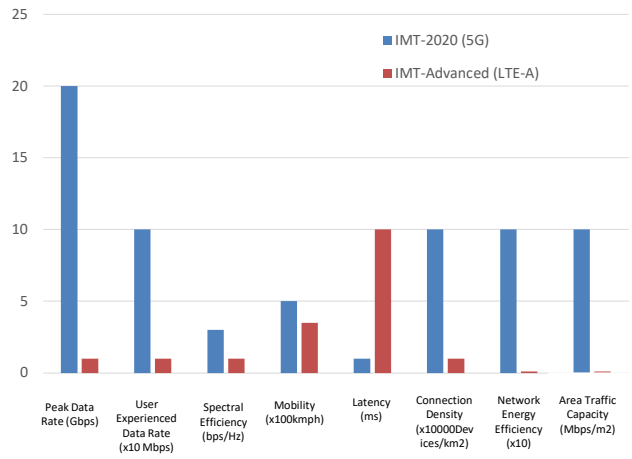
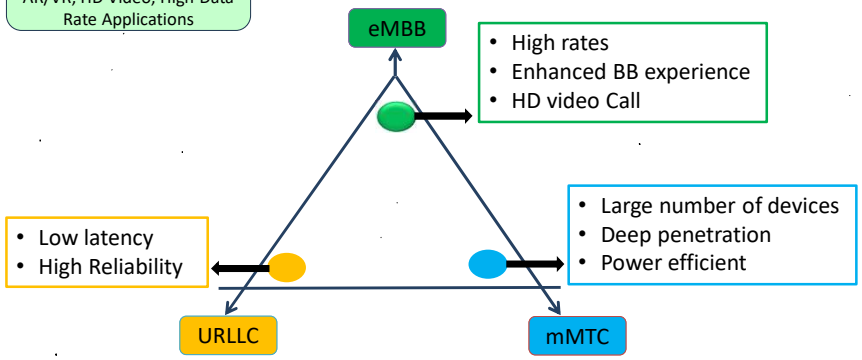
5G Performance Requirements(ITU)



5G Service Categories

Fixed Wireless Access (FWA); FTTH alternative, AR/VR, HD Video, High Data Rate Applications

- Enhanced Mobile Broad-Band



- High rates
- Enhanced BB experience
- HD video Call

- Low latency
- High Reliability

- Large number of devices
- Deep penetration
- Power efficient

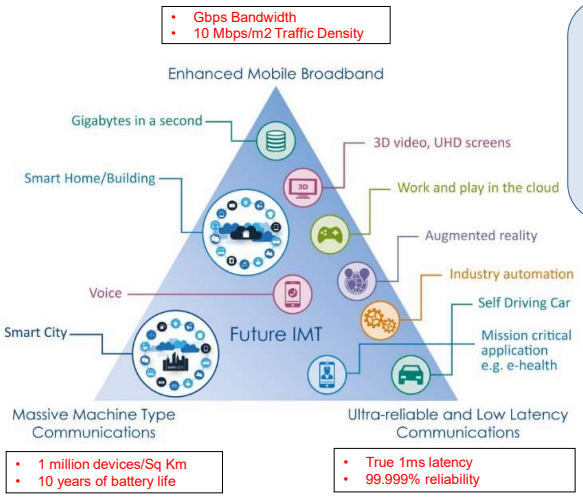
- Ultra-Reliable and Low Latency Communication

Vehicle to Vehicle and others, Mission Critical Communication, Factory Automation, etc

- Massive Machine Type Communication

Smart Home, Smart City, Tracking, Fleet Management, Smart Meter, etc

5G Vision

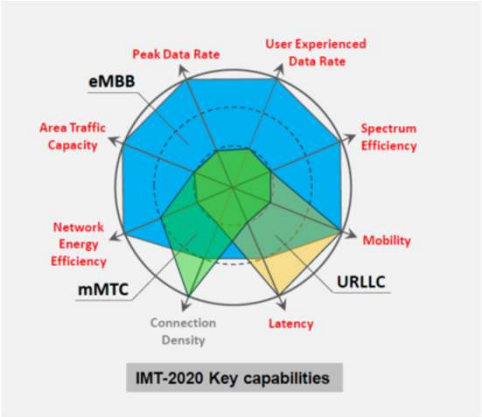


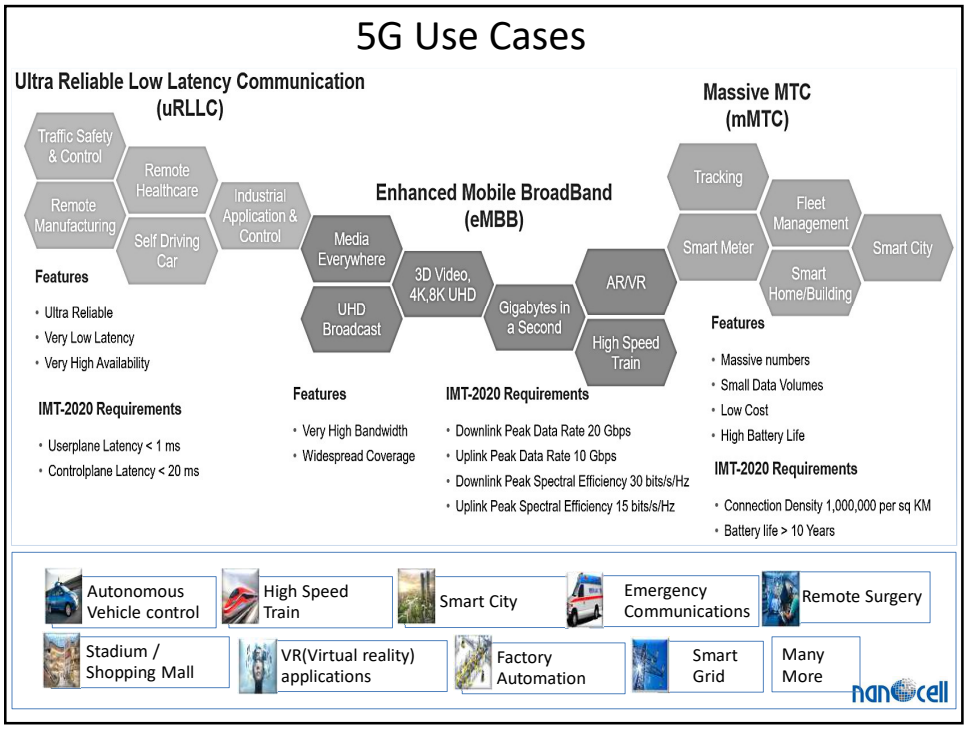
- Low Latency Radio
- Mission Critical Applications
- Industry 4.0 Integration
- Ready for AR/VR applications
- Ready for AI/ML to be used in the network
- Connect everything everywhere

Source: 3GPP

Services and Requirements Characteristics

Characteristics	mMTC	URLLC	eMBB
Availability	Regular	Very High	Regular (baseline)
E2E latency	Not highly sensitive	Extremely sensitive	Not highly sensitive
Throughput type	Low	Low/med/high	Medium
Frequency of Xfers	Low	High	High
Density	High	Medium	High
Network coverage	Full	Localized	Full





Some Use Cases & Requirements

Use Cases	Requirements	Desired
Autonomous vehicle control	Latency Availability	5ms 99.999%
High Speed Train	Traffic volume density Experienced throughput	50-100 Gbps/sq.km 25-50 Mbps
Factory automation	Latency Availability	Less than 1ms 99.99%
Massive amount of devices	Connection density Availability Energy efficiency	1 million per km2 99.9 % coverage 10 Years of Battery
VR applications	Experienced Throughput Latency	Gbps 10ms
Remote Surgery	Latency Reliability	<1ms 99.9%

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Use cases from 5G Connected Industries Alliance

Use case (high level)		Availability	Cycle time	Typical payload size	# of devices	Typical service area
Motion control	Printing machine	>99.9999%	< 2 ms	20 bytes	>100	100 m x 100 m x 30 m
	Machine tool	>99.9999%	< 0.5 ms	50 bytes	~20	15 m x 15 m x 3 m
	Packaging machine	>99.9999%	< 1 ms	40 bytes	~50	10 m x 5 m x 3 m
Mobile robots	Cooperative motion control	>99.9999%	1 ms	40-250 bytes	100	< 1 km²
	Video-operated remote control	>99.9999%	10 – 100 ms	15 – 150 kbytes	100	< 1 km²
Mobile control panels with safety functions	Assembly robots or milling machines	>99.9999%	4-8 ms	40-250 bytes	4	10 m x 10 m
	Mobile cranes	>99.9999%	12 ms	40-250 bytes	2	40 m x 60 m
Process automation (process monitoring)		>99.99%	> 50 ms	Varies	10000 devices per km²	

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5G Spectrum – Frequency Ranges

Two Frequency ranges for 5G

- Frequency Range1 - FR1 (450 MHz -7.125 GHz)
  - Bands numbered from 1 to 255
  - No longer can be commonly referred to as sub-6GHz
  - FR1 has both FDD and TDD bands
- Frequency Range2 - FR2-1 (24.250 -52.600 GHz) - May get extended
  - Bands numbered from 257 to 511
  - Commonly referred to as mm Wave
  - FR2 bands are TDD bands

Multiple disparate spectrum regions could be deployed for 5G

Different applications, different deployments are the drivers for a wide spectrum range to be considered

mm-wave high band spectrum is the new frontier for 5G

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5G Spectrum

- Channel Bandwidth in FR1 varies from 5MHz to 100MHz
  5, 10, 15, 20, 25, 30, 40, 50, 60, 80, 100MHz
- FR1 is suited for medium capacity good coverage 5G Network
- Channel Bandwidth in FR2 varies from 50MHz to 400MHz
  50, 100, 200, 400MHz
- Fixed and Mobile Access likely on FR2
- FR2 is suited for hotspots and capacity management
- Beamforming to improve FR2 performance

5G Frequency Bands

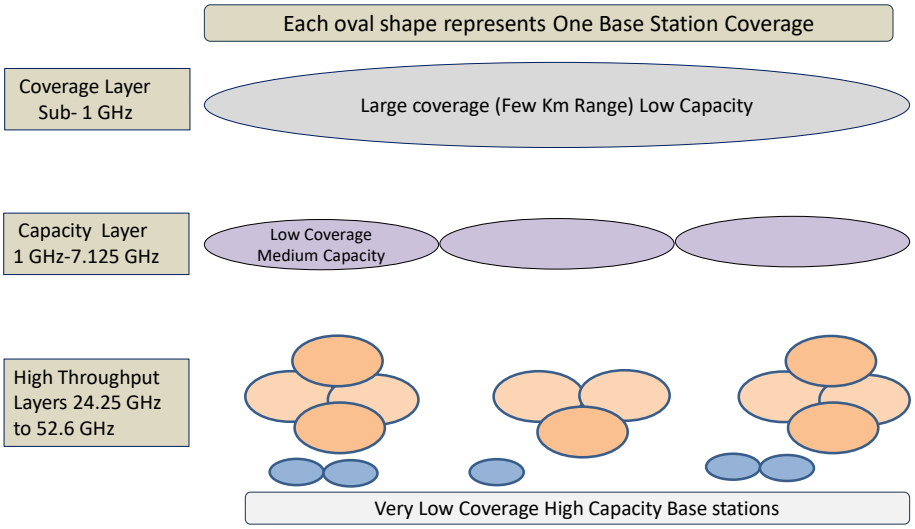
FR1 – Below 7.2 GHz; LTE bands can also be used for 5G

All new bands in FR1 are slated for TDD operation in R 15 but FDD operation in older bands is possible
5G NR new bands in NR slated to support larger channel bandwidths

FR2 ; 24-52 GHz; "mm-wave spectrum" TDD only operations; Large bandwidth channels; likely to be used for fixed access to start with and then mobile

Source: 3gpp

5G Multiple Coverage Layers For Multiple Needs



Spectrum needs for 5G Service Categories

eMBB

- New spectrum needed
- Both FR1 and FR2 to be used. FR2 has large carrier bandwidth
- Large contiguous channel BW needed
- Coverage, Unlicensed bands

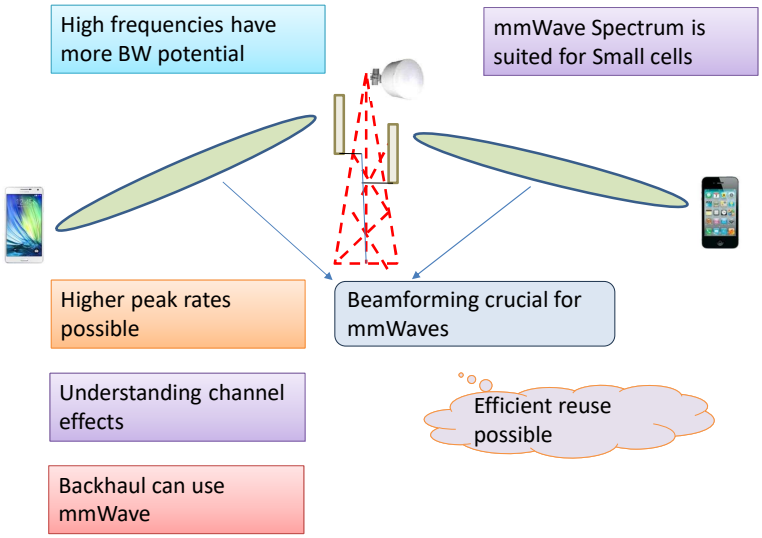
mMTC

- Sub 1GHz preffered
- Small channel BW Ok
- Stable licensed spectrum

uRLLC

- Exclusive or high priority access needed
- Use of frequency diversity for reliablity

mmWave systems



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5G Network Overview

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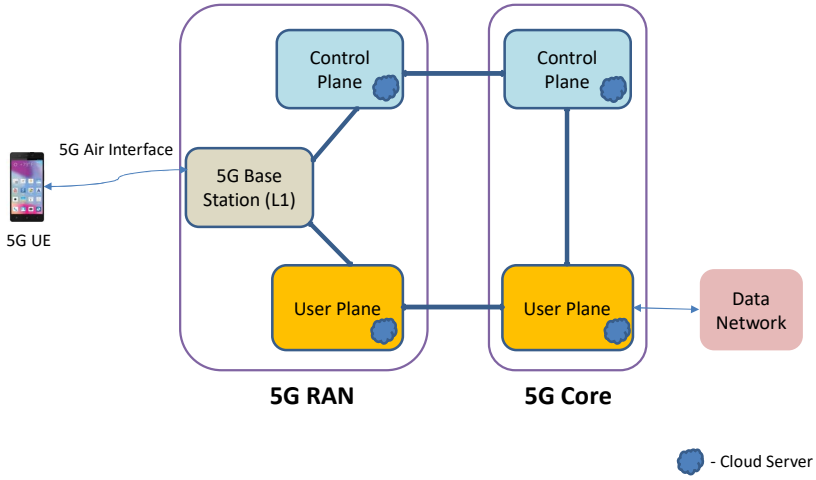
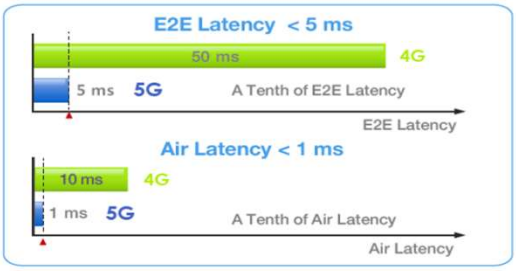
5G Network Architecture – Broad Approach

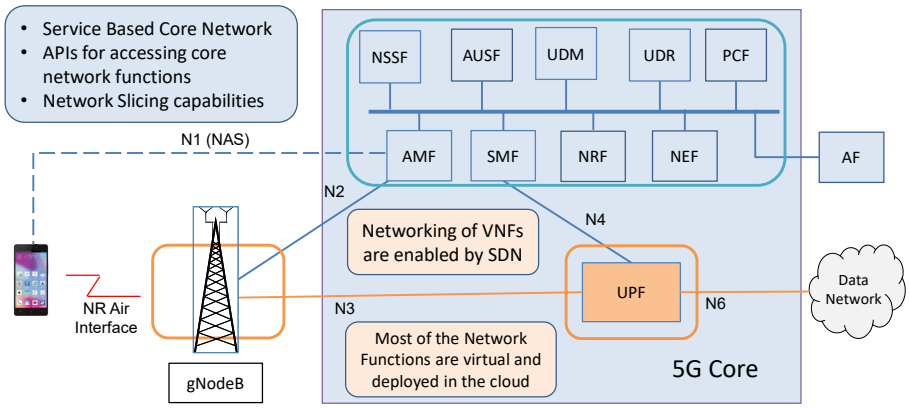


How to achieve low Latency in Cellular Network



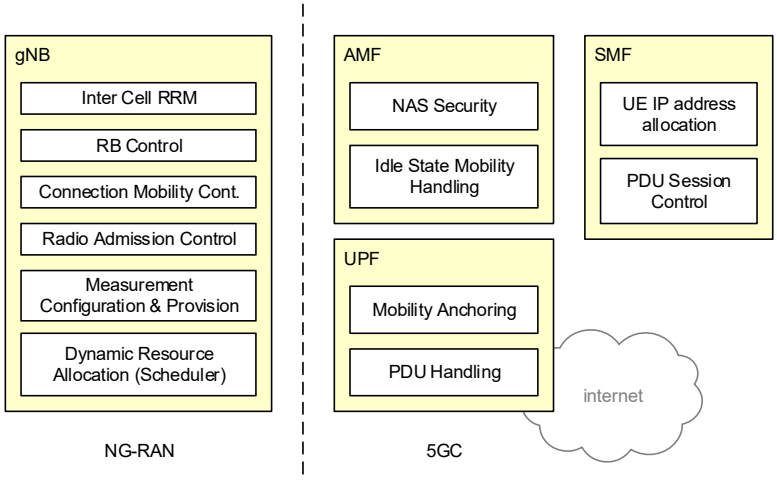
- Separation of C & U plane in Network
- Flat Network Architecture
- Faster Processing
- Smaller TTI
- Better Design of Network & Implementation

5G Network Architecture



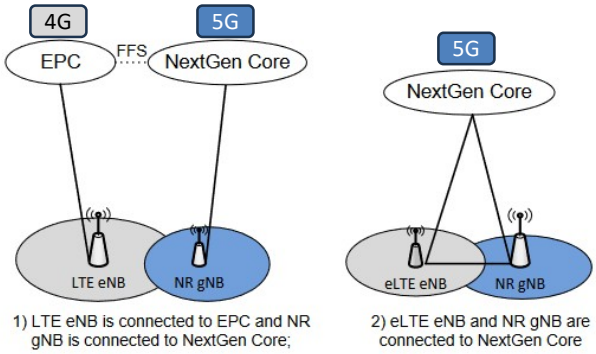
AMF	Access & Mobility Management Function	**NSSF**	NW Slice Selection Function	**SMF**	Session Management Function	**UDSF**	Unstructured Data Storage Function
AUSF	Authentication Server Function	**NEF**	Network Exposure Function	**UPF**	User Plane Function	**NWDAF**	Network Data Analytics Function
PCF	Policy Control Function	**NRF**	Network Repository Function	**UDM** **UDR**	Unified Data Management User Data Repository	**AF**	Application Function

Role of 5G RAN and a few Core Network Functions



Source: 3GPP

Introduction of the NG-RAN: gNodeB

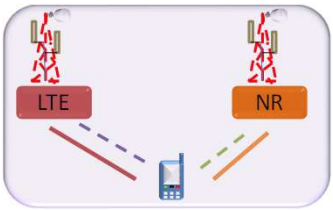


5G specifications are ongoing. The base station in 5G is called **gNB**

The new radio access technology is called **"NR"** (New Radio)

5G gNB can also be integrated with 4G RAN and / or EPC

5G RAN Approach



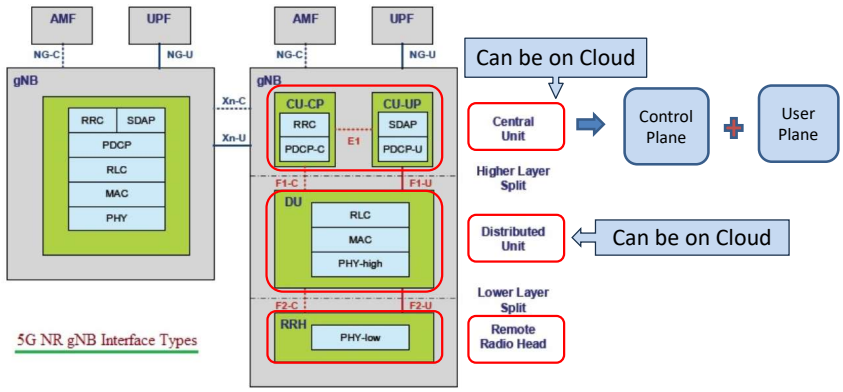
5G is evolving and many features will come in future releases to complete the 5G picture

Rel. 15 Phase 1 RAN specifications addressed only eMBB with advancements towards the low-latency part of URLLC

Only NSA approach in phase 1, i.e., always with LTE as the anchor

Rel. 15 Phase 2 RAN specifications addressed standalone 5G NR operations including coverage of the robustness part for URLLC

gNodeB – Overview

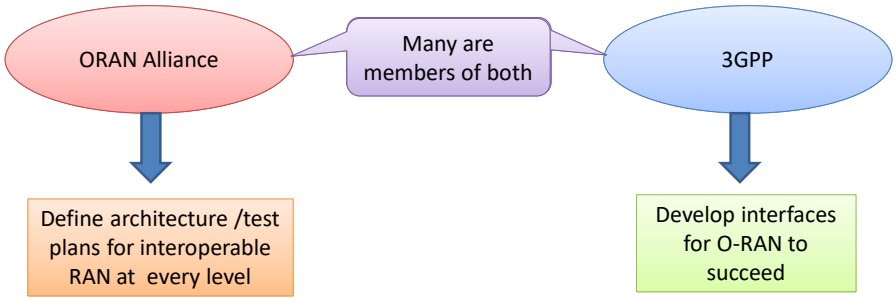


Picture Source: rcf-wireless

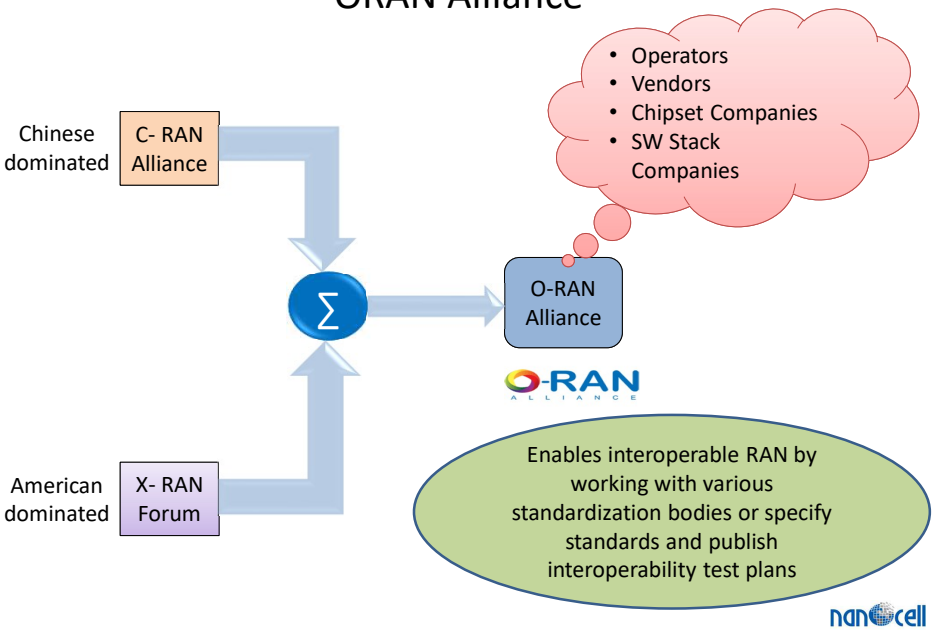
Why Open RAN

- Reduce Dependency on RAN Vendors / No vendor Lock-in
- Promote Innovation
- Software Centric Networks
- Virtualization of RAN to get benefits of virtualization
- Quick Time-To-Market for new services
- Flexibility to Implement RAN

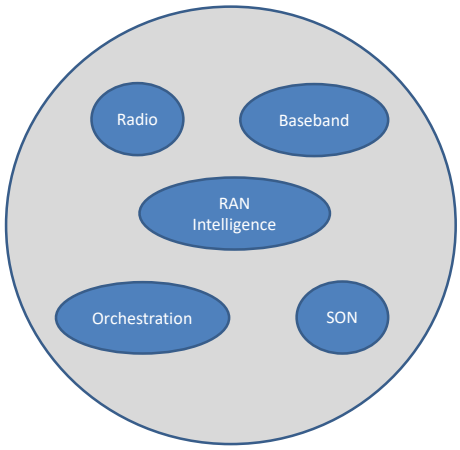
Who are involved in Open RAN?



ORAN Alliance

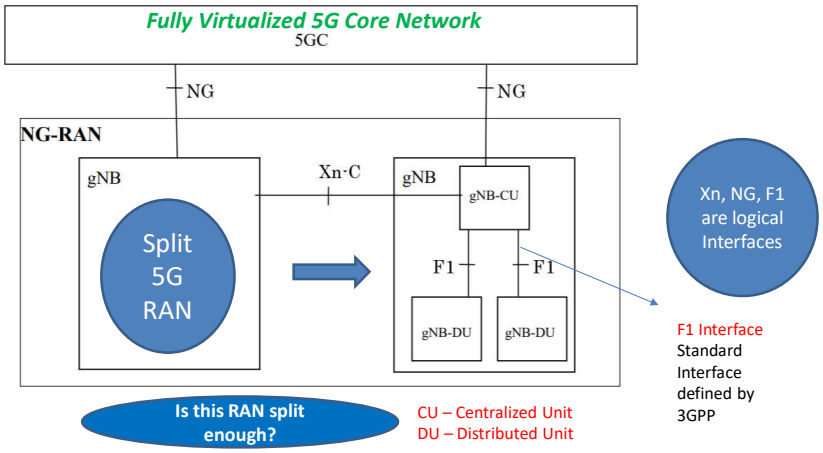


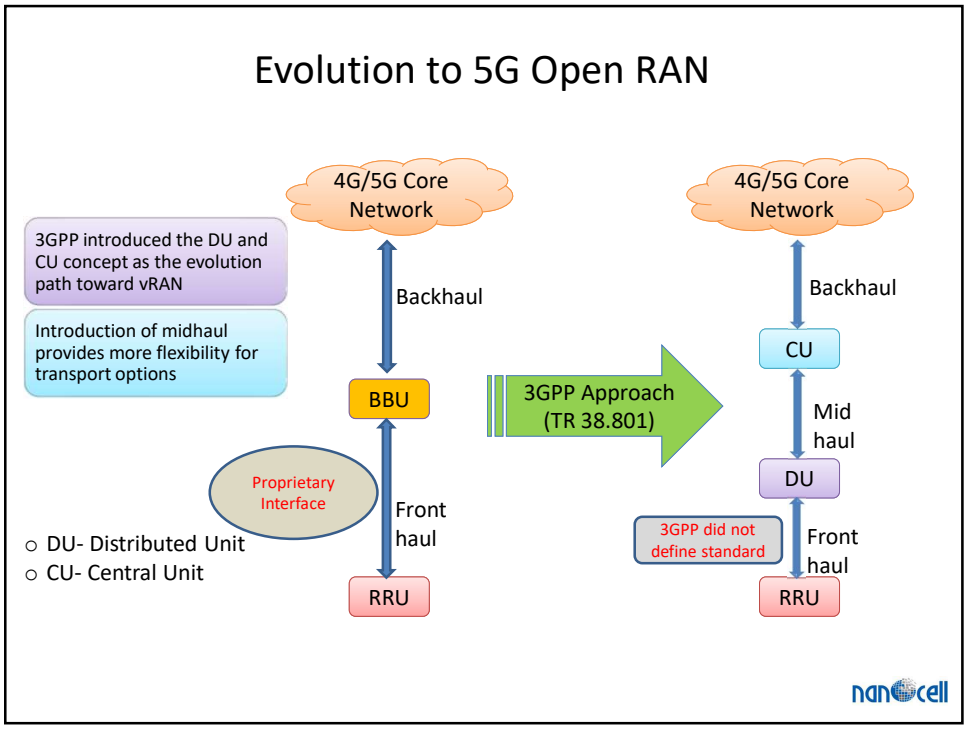
ORAN and 3GPP efforts to Open the RAN



If we open the RAN, can we get each of this from different supplier?

Overall Architecture of NG-RAN




Evolution to 5G Open RAN
4G/5G Core Network
3GPP introduced the DU and CU concept as the evolution path toward vRAN
Introduction of midhaul provides more flexibility for transport options
Backhaul
BBU
Proprietary Interface
Front haul
RRU
DU- Distributed Unit
CU- Central Unit
3GPP Approach (TR 38.801)
4G/5G Core Network
Backhaul
CU
Mid haul
DU
3GPP did not define standard
Front haul
RRU
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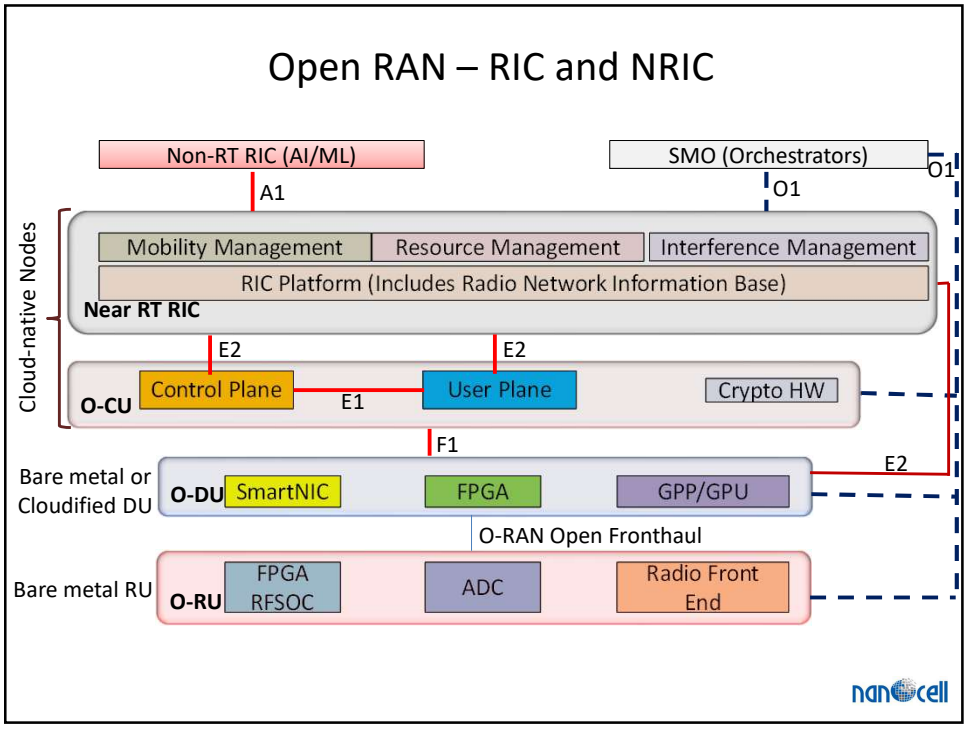
Open RAN – RIC and NRIC
Non-RT RIC (AI/ML)
SMO (Orchestrators)
O1
A1
O1
Cloud-native Nodes
Mobility Management
Resource Management
Interference Management
RIC Platform (Includes Radio Network Information Base)
Near RT RIC
E2
E2
O-CU
Control Plane
E1
User Plane
Crypto HW
F1
E2
Bare metal or Cloudified DU
O-DU
SmartNIC
FPGA
GPP/GPU
O-RAN Open Fronthaul
Bare metal RU
O-RU
FPGA RFSOC
ADC
Radio Front End
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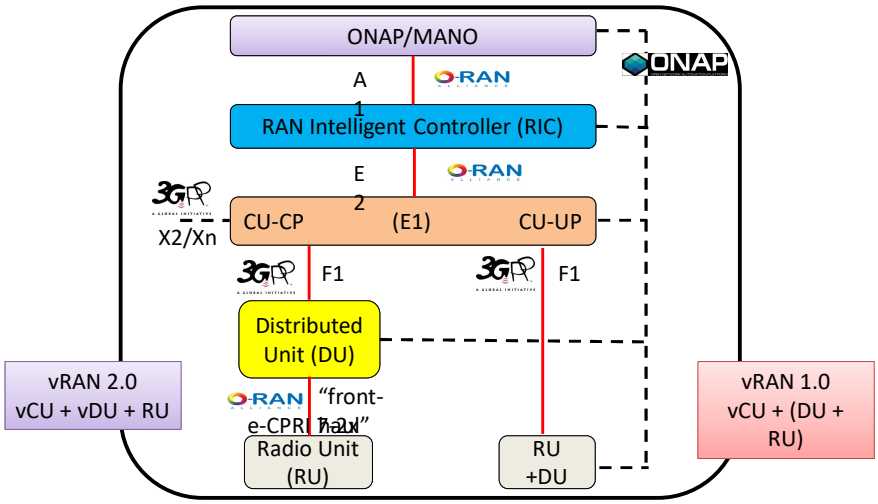
RAN Functions & Interfaces – 3GPP and O-RAN alliance



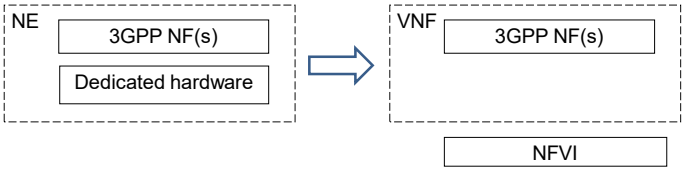
Source: Nokia

5G Core Network – Architecture Principles

- Clear separation of UP and CP
- Make the functions granular
- Virtualize functions
- Flexible placement of functions
- Elasticity: Scalable to meet SLAs. Movable to other servers.
- Resiliency: Be able to recreate after failure
- Service Continuity: Seamless or non-seamless continuity after failures or migration
- Service Assurance: Time stamp and forward copies of packets for Fault detection
- Energy Efficiency Requirements: Should be possible to put a subset of VNF in a power conserving sleep state
- Access agnostic

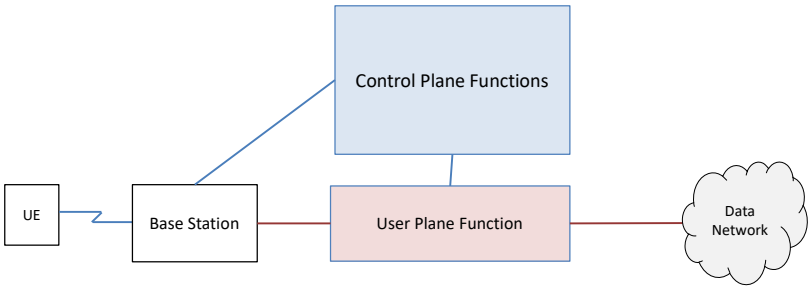
Principle 1: Virtualise the Functions

- Network Function Virtualization (NFV)
  - Decouple Network Function from hardware
    - Develop as Virtual Network Functions i.e software functions
    - COTS hardware
  - Stateless, Scale on demand
  - Design to run on a cloud (VMs / Containers etc)



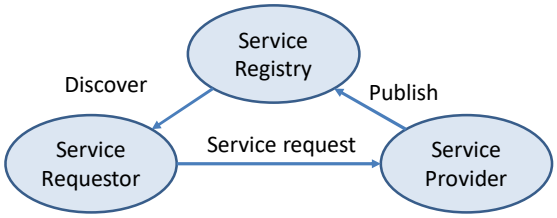
Principle 2: Control and User Plane Separation

- Control User Plane Separation
  - Control functions totally separated from Data plane functions
  - Independent scaling and provisioning
  - Data plane can move closer to user for low latency

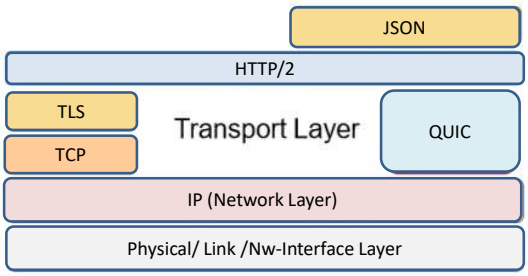


Principle 3: Service Based Architecture

- Service Based Architecture (SBA)
  - Control plane components talk over HTTP
  - Network Functions as a suite of Services - RESTful APIs
- Making the Network Function Stateless
- Protocol based signaling -> API/Service based signaling



Service Based Architecture - Stack and Interfaces



- Representational State Transfer (REST) API based communication
- HTTP/2 Protocol – Commands Like PUT/GET/DELETE etc
- Uses HTTP protocol to transport JSON format messages in REST API style
- HTTP/2 (originally named HTTP/2.0) is major revision of the HTTP network protocol used by the World Wide Web. It was derived from the earlier experimental SDPY protocol, originally developed by Google
- HTTP/2 is binary, instead of textual.
- HTTP/2 is fully multiplexed

5G Core Functions

Access and Mobility Management function (AMF) supports: Termination of NAS signalling, NAS ciphering & integrity protection, registration management, connection management, mobility management, access authentication and authorization, security context management

Session Management Function (SMF) supports: session management (session establishment, modification, release), UE IP address allocation & management, DHCP functions, termination of NAS signalling related to session management, DL data notification, traffic steering configuration for UPF for proper traffic routing.

Authentication Server Function (AUSF) acts as an authentication server.

Unified Data Management (UDM) supports: generation of Authentication and Key Agreement (AKA) credentials, user identification handling, access authorization, subscription management.

User Data Repository (UDR) supports: Storage and retrieval of subscription, policy data of subscribers and application data

User Plane Function (UPF) supports: packet routing & forwarding, packet inspection, QoS handling, acts as external PDU session point of interconnect to Data Network (DN), and is an anchor point for intra- & inter-RAT mobility.

Application Function (AF) supports: application influence on traffic routing, accessing NEF, interaction with policy framework for policy control

Network Exposure function (NEF) supports: exposure of capabilities and events, secure provision of information from external application to 3GPP network, translation of internal/external information

NF Repository function (NRF) supports: service discovery function, maintains NF profile and available NF instances

Network Slice Selection Function (NSSF) supports: selecting of the Network Slice instances to serve the UE, determining the allowed NSSAI, determining the AMF set to be used to serve the UE

Similar to 4G Core Functions

New in 5G

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Key System Procedures involving 5G Core

Connection, Registration and Mobility Management procedures

- Registration management
  - Authentication
  - Security mode
  - UE context establishment
- UE Configuration Update
- UE Mobility Management

Policy Management
Billing
Legal Intercept
Quality of Service
Switching and Routing

Session Management procedures

- Data Sessions for carrying data traffic
  - DNNs -> PDU Sessions -> QoS Flows -> IP Traffic
  - PDU Session types: IPv4, IPv6, Ethernet, Unstructured
  - UE can establish the PDU session over any AN available
- PDU Session Establishment/Modification/Release

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OSS and BSS in 5G

Operation Support System (OSS)

Overall Objective is to manage the operations of the 5G network.

Key Functions:

- **Network Monitoring:** Real-time tracking of network performance and infrastructure.
- **Fault Management:** Identifying and resolving network issues and failures.
- **Configuration:** Configuration of network Nodes/Functions and their settings
- **Inventory Management:** Keeping track of network resources
- **Capacity Planning:** Optimizing network capacity to manage demand

Business Support System (BSS)

Customer-facing and business process management

Key Functions:

- **Billing:** Generating invoices for post paid customers.
- **Customer Relationship Management (CRM):** Managing customer interactions and data.
- **Order Management:** Handling customer orders for services.
- **Product and Service Creation:** Developing and offering new 5G services to customers.
- **Settlements:** Managing inter-business transactions.
- IN – Intelligent Network (PCRF, PCF), Manage Pre-Paid Customers

OSS and BSS in 5G

How 5G OSS / BSS is Implemented

1. Cloud-Native Architecture: Offers flexibility, scalability, and agility

2. Automation: Manage complex processes like service provisioning and fault resolution to support 5G's high volumes of data and service types

3. Network Slicing: Management of virtual, tailored network slices for different services, a key capability of 5G.

4. Monetization of Network Assets and Services: Enables operator to monetize by creating and delivering new, specialized services to businesses and consumers

5. Artificial Intelligence (AI) and Machine Learning (ML): Key in integrated OSS/BSS to improve network insights, automate complex decisions, and enhance the overall customer experience.

Key System Procedures involving 5G Core

Connection, Registration and Mobility Management procedures

- Registration management
  - Authentication
  - Security mode
  - UE context establishment
- UE Configuration Update
- UE Mobility Management

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QoS in 5G

Introduction

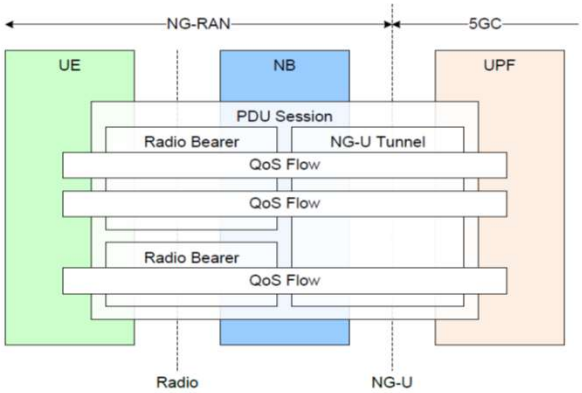
- PDU Session: It provides end-to-end user plane connectivity between the UE and a specific Data Network (DN) through the User Plane Function (UPF).
- Within a PDU Session, there can be one or more QoS Flows (to support different applications for the same UE)
- QoS Flow and QoS profile is mapped one to one, i.e. all packets belonging to a specific QoS Flow will have same "5G Quality of Service Identifier" (5QI)

Need for QoS
- Different Subscription levels
- Differentiating service types – e.g. voice, net browsing
- Different Application requirements
- Use cases – eMBB, uRLLC, IOT

QoS Implementation
- Assigning radio, network, and transport resources
- Allocation and Retention priority
- Pre-empt the existing traffic
- Checkpoint for traffic

QoS Flows in 5G

- QoS Flow is the finest granularity of QoS differentiation in the PDU Session.
- A QoS Flow ID (QFI) is used to identify a QoS Flow in the 5G System.



QoS Profile

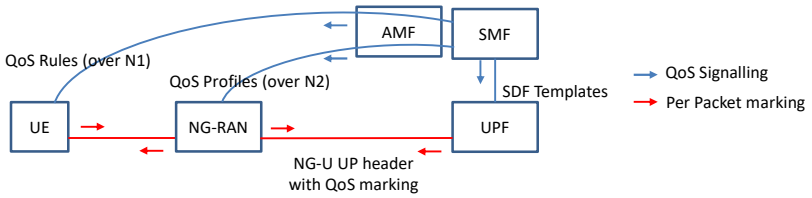
- 5G QoS Profile defines the Quality of Service (QoS) characteristics for a data flow, ensuring specific service requirements for applications like voice, video, or IoT
- Associated with a 5G QoS Identifier (5QI) and contains parameters like packet delay budget, packet error loss, and bit rates
- The profile determines how packets are handled by mapping them to a QoS Flow

QoS Identities	GBR	Non GBR
• 5G QoS Identifier (5QI); • Allocation and Retention Priority (ARP) • QFI – QoS Flow Identifier	• Guaranteed Flow Bit Rate (GFBR) - UL and DL; and • Maximum Flow Bit Rate (MFBR) - UL and DL • Notification control • Maximum Packet Loss Rate - UL and DL	• Reflective QoS Attribute (RQA)

5QI Values

5QI Value	Resource Type	Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume (NOTE 2)	Example Services
B	Delay Critical GBR	11	5 ms	10^{-5}	160 B	Remote control (see TS 22.261 [2])
C NOTE 4		12	10 ms NOTE 5	10^{-6}	320 B	Intelligent transport systems
D		13	20 ms	10^{-5}	640 B	
1	GBR NOTE 1	20	100 ms	10^{-2}	N/A	Conversational Voice
2		40	150 ms	10^{-3}	N/A	Conversational Video (Live Streaming)
3		30	50 ms	10^{-3}	N/A	Real Time Gaming, V2X messages Electricity distribution – medium voltage, Process automation - monitoring
4		50	300 ms	10^{-6}	N/A	Non-Conversational Video (Buffered Streaming)
65		7	75 ms	10^{-2}	N/A	Mission Critical user plane Push To Talk voice (e.g., MCPTT)
66		20	100 ms	10^{-2}	N/A	Non-Mission-Critical user plane Push To Talk voice
75		25	50 ms	10^{-2}	N/A	V2X messages
E NOTE 4		18	10 ms	10^{-6}	255 B	Discrete Automation
F NOTE 4		19	10 ms	10^{-6}	1358 B NOTE 3	Discrete Automation
5	Non-GBR NOTE 1	10	100 ms	10^{-6}		IMS Signalling
6		60	300 ms	10^{-6}		Video (Buffered Streaming) TCP-based (e.g., www, e-mail, chat, ftp, p2p file sharing, progressive video, etc.)
7		70	100 ms	10^{-3}		Voice, Video (Live Streaming) Interactive Gaming
8		80	300 ms	10^{-6}		Video (Buffered Streaming) TCP-based (e.g., www, e-mail, chat, ftp, p2p file
9		90				sharing, progressive video, etc.)
69		5	60 ms	10^{-6}		Mission Critical delay sensitive signalling (e.g., MC-PTT signalling)
70		55	200 ms	10^{-6}		Mission Critical Data (e.g. example services are the same as QCI 6/8/9)
79		65	50 ms	10^{-2}		V2X messages
G		66	10 ms	10^{-6}	N/A	Low Latency eMBB applications Augmented Reality

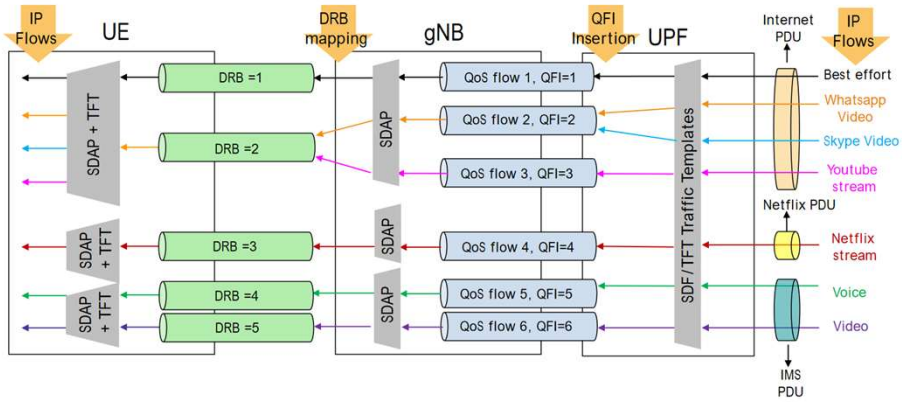
QoS framework



Any QoS Flow is characterised by

- A QoS profile provided by the SMF or preconfigured
- QoS rule(s) provided by the SMF to the UE and/or derived by the UE by applying **Reflective QoS control**
- One or more SDF (Service Data Flow) templates provided by the SMF to the UPF

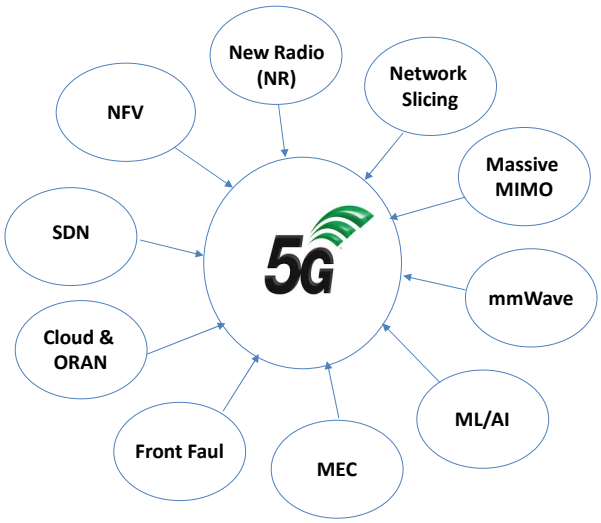
5G QoS Flow example in downlink



Source: Internet – Devopedia.org

5G Network – Key Features & Enablers

5G – Enablers & Technologies



5G – Enablers & Technologies

New Radio	Flexibility to handle multiple use case scenarios in air interface
Network Slicing	Virtual Mobile Network to address multiple services
mmWave	New Frequency Range to address high Capacity
Massive MIMO	To manage capacity and coverage (using beamforming)
ORAN / CRAN	RAN is split into multiple functional units with open interfaces
SDN	Software Defined Networking with centralized management
Cloud	Scale-up and Scale-Down Network Capacity with SDN/NFV
Virtualization	To disintegrate Network Software from hardware to enable
AI / ML	To enable Autonomous Networks with SON / Zero Touch

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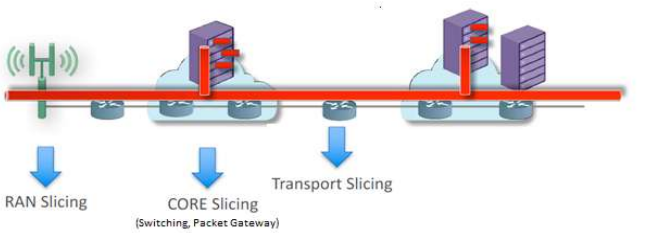
Network Slicing

Network slicing is a mechanism to create and dynamically manage functionally-discrete, virtualized networks over a common infrastructure.



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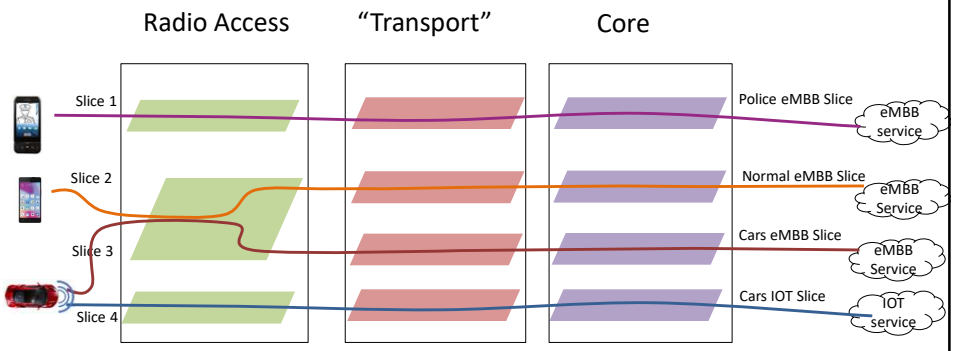
Network Slicing Scenarios

- Three fundamental scenarios have been identified for 5G: Service Slice Type (SST)
  - eMBB : 5G enhanced Mobile Broadband.
    - Video surveillance and mobile medical services express high requirements on the transmission rate
  - MIoT : Massive IoT.
    - For intelligent city, smart grid, intelligent agriculture and other services, networks need to support massive equipment access sending small data.
    - IoT services need different types of networks and requirements for mobility, billing, security, policy control, latency and reliability.
  - URLLC : Ultra- reliable low latency communication
    - Automotive networking and vehicle to everything (V2X) communication and industrial control services require nearly 100% reliability and millisecond latency
- There are many use cases under each service catergory. Each use case may be served by a network slice to meet the performance requirements on 5G network
- User specific slices apart from these 3 can also be configured with non-standard SST values.

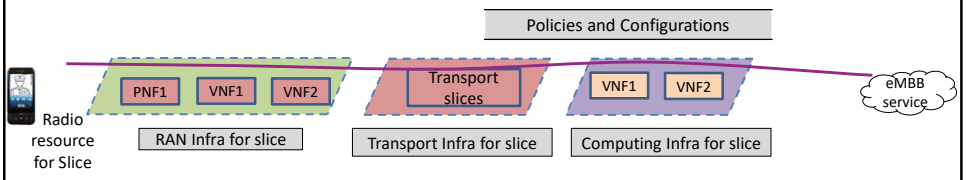
End to end Slice through RAN, Core and Transport



- A UE can access multiple slices in the Network
- A Slice can service multiple UEs and Services
- A Slice is realised at RAN, Core and Transport

What is a Network Slice Composition

- A set of network functions
- Resources to run these network functions
- Policies and configurations
- Specific configurations of transport network
- Dedicated radio configuration or even a specific RAT
- Configuration of the 5G device



Benefits of Network Slicing

- Widens the Customer Base for the Operators / Infrastructure Providers
  - Slices for Vertical Applications
  - Slices for diverse Horizontal services
  - Non-public networks
- New business models
  - Ex. Virtual Networks for virtual operators
- Optimising of network resources
  - Right Sizing
- Network as a service
  - On demand slices!
- And More as the concept evolves

ML & AI in 5G

Few areas of AI/ML Implementation
Scheduling Beamforming in Massive MIMO Networks
Indoor Positioning of devices
Configuration of Uplink and Downlink Channels
Real-time Optimization of Network, RRM, Automatic Neighbour Management, Handover Optimization, etc
Service Provisioning, Subscriber Behaviour & Profiling

- Implementation of AI in Networks will be done in phase wise
- In the network AI/ML Implementation can be
  - EDGE
  - Distributed
  - Centralized

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MEC

- Multi-Access Edge Computing (MEC)
  - moving **'compute'** resource
    - User data
  - to the **'edge'**
    - Physically close to the user
    - Typically, close to the access network
    - Can also be placed at enterprise premises, e.g. inside factory buildings, offices, homes, vehicles etc.
  - **Multi-Access**
    - Not just 'mobile' anymore – variety of devices and access types
- Why
  - Low latency
  - Higher reliability (shorter path + typically tighter control over path)
  - High bandwidth
  - Optimized device data processing
  - Data offload (reduce backhaul traffic by keeping right content at edge)
  - Trusted computing and storage

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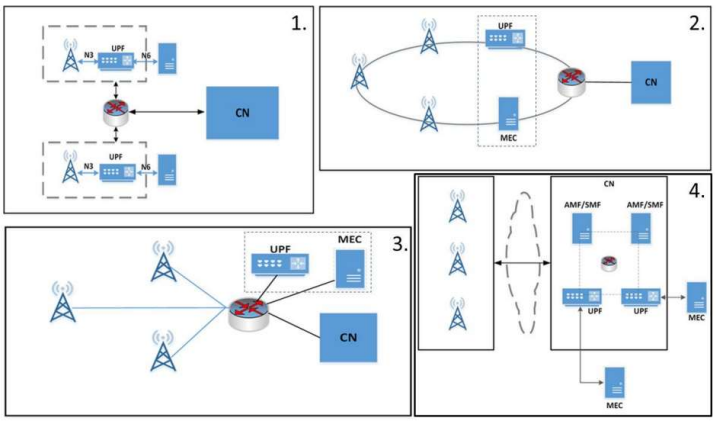
MEC Deployment Scenario



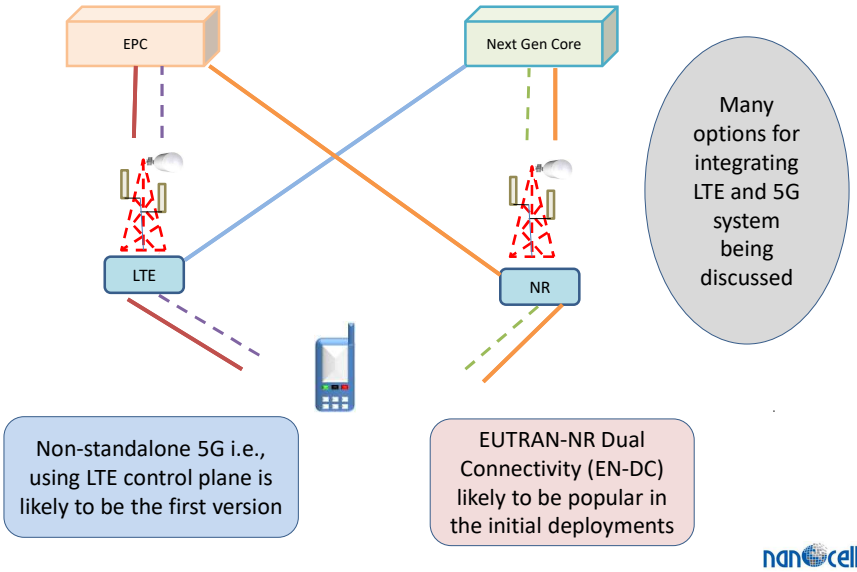
Examples of the physical deployment of MEC.

1. MEC and the local UPF collocated with the Base Station.
2. MEC collocated with a transmission node, possibly with a local UPF
3. MEC and the local UPF collocated with a network aggregation point
4. MEC collocated with the Core Network functions (i.e. in the same data centre)

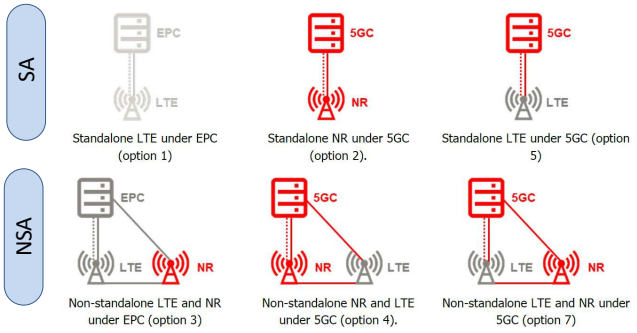
Source: ETSI White Paper

5G Deployment & Standards Update

5G Deployment – Network Architecture Options



5G Deployment Options



- There are mainly six deployment options in both SA and NSA modes.
- Option-5 using 5GC and LTE ng-eNodeB access
- Option-3 using EPC and LTE eNodeB acting as master and NR eNodeB-gNodeB acting as secondary;
- Option-4 using 5GC and an NR gNodeB acting as master and LTE ng-eNodeB acting as secondary

- Option-7 using 5GC and an LTE ng-eNodeB acting as master and an NR gNodeB acting as secondary.

Source: GSMA

What should an operator do to implement 5G?

5G Spectrum	Can operator get low-band 5G NR Spectrum to get good coverage?
5G RAN	MIMO Configuration, Channel Bandwidth, NSA / SA options
5G Network Architecture	Business / Technical considerations for rolling out a 5G CN, MEC? SA or NSA?
eLTE	Does it make sense to upgrade to eLTE? Will devices be available?
5G Service	What are the different 5G Services to be offered
Spectrum refarming	When can 2G/3G be retired and spectrum be used for 5G?

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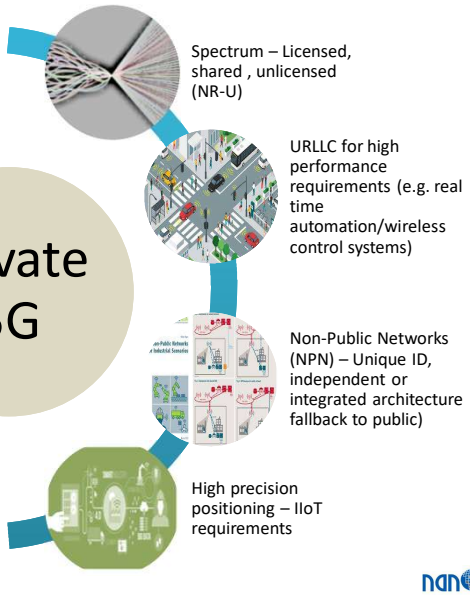
Private 5G : Dedicated 3GPP features for Industry 4.0 and 5.0 Deployment



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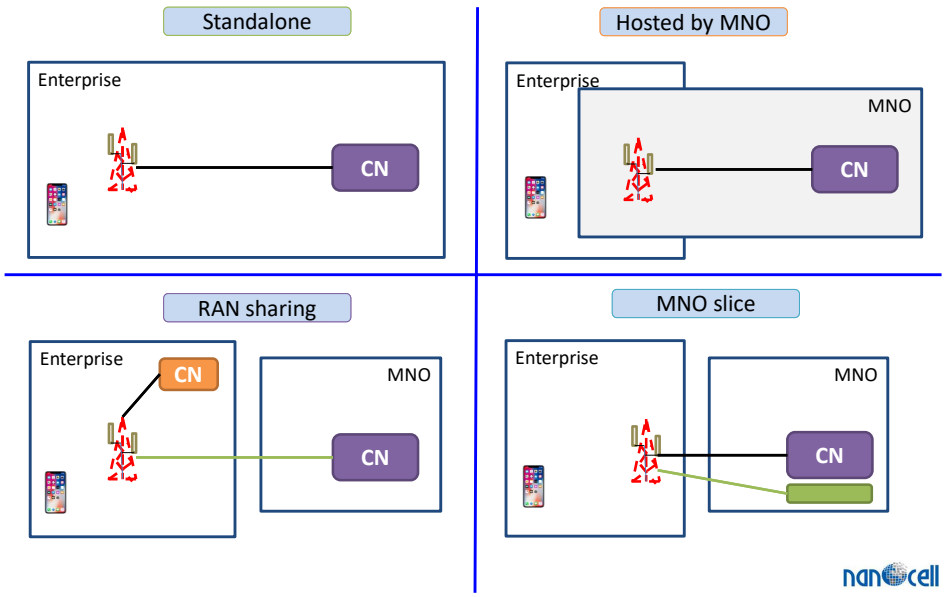
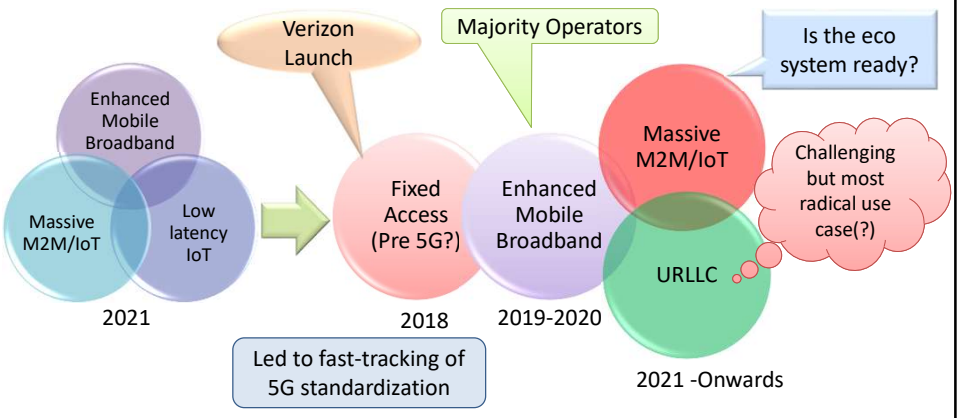
Private 5G Networks – High Level Implementation Options



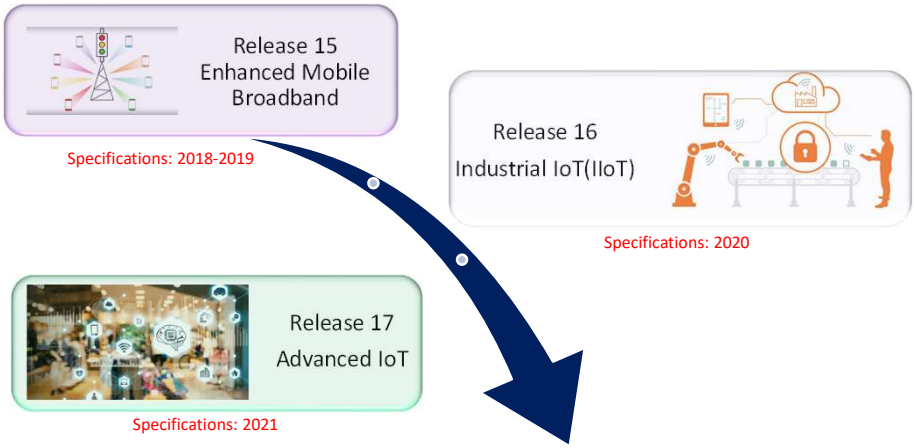
5G Deployment Evolution – Global Scenario



As on Sept 2025 (Source: Online, Ericsson Mobility Report)

- About 300 MNOs on 5G across the world offering 5G Services and FWA
- Gradual deployment of 5G Private Networks by various industry verticals, Estimated about 4500 Networks are deployed
- Mix of NSA and SA deployments
- 2.5 to 3 billion users on 5G is estimated

5G Evolution – Big Picture



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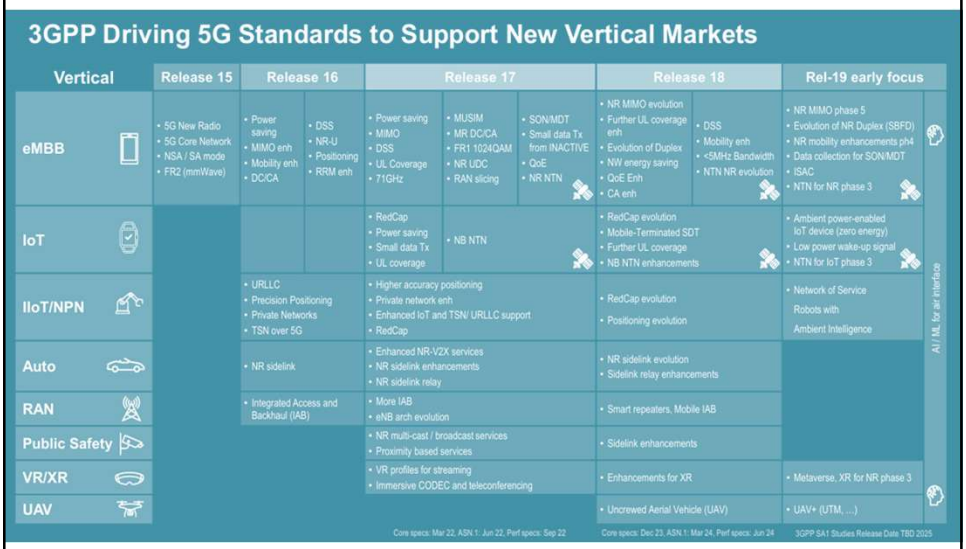
5G Releases - Impact to other Industries



Picture Source: Keysight

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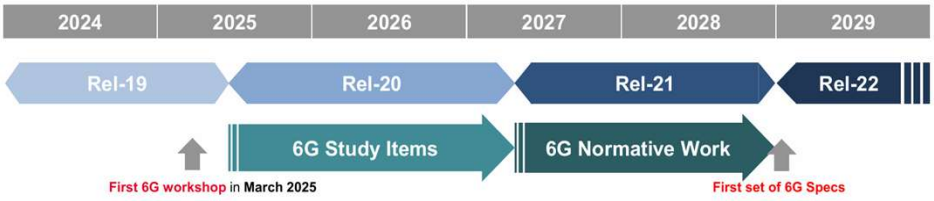
5G Evolution to 6G – Vision & Timelines



IMT-2030 6G Vision

- Ubiquitous Connectivity and Intelligence
- Integration of Communication and Sensing
- Seamless Coverage
- Sustainability and Resilience
- Security and Privacy
- AI-Native Networks

Bharat 6G Vision 2030

- Indigenous Design and Development
- Key Pillars: Affordability, Sustainability, and Ubiquity.
- Ecosystem Development for research
- Global Leadership

Picture Source: Keysight

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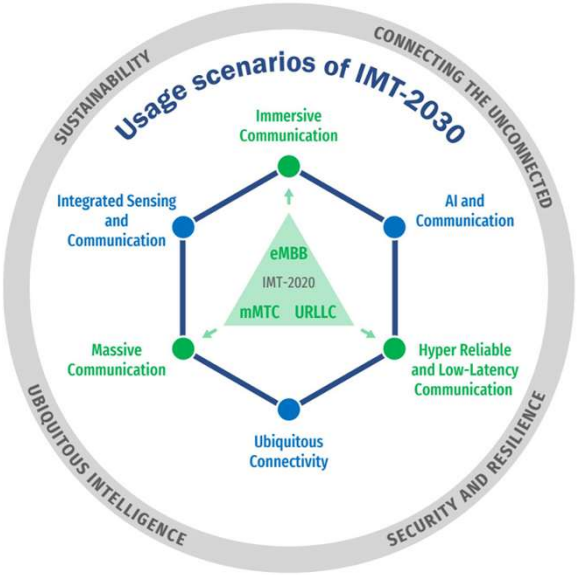
6G Expectations



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6G Performance Requirements

Capacity

Data Rates: Terabits per second (Tbps), significantly faster than 5G.
Latency: Sub-millisecond (microsecond) end-to-end latency.
Reliability: Extremely low jitter and high reliability for critical applications.
Connection Density: Over 1 million devices per square kilometer.

Coverage & Mobility

Ubiquitous 3D Coverage: Seamless integration of terrestrial, aerial, and satellite networks.
High Mobility: Support for speeds up to 500 km/h or more, extending 5G's capabilities.

Intelligence & Design

AI-Native: AI built into the core for massive automation and network optimization.
Sustainable: Energy-efficient design and operations.
Open & Converged: Open RAN architecture and federated control for service composition.

Key Technologies & Capabilities

Spectrum: Utilization of new mid-band (4-15 GHz) and high-frequency (THz) spectrum.
Sensing: Integrated sensing and positioning capabilities.
Security: Advanced security, privacy, and trustworthiness frameworks.

Enabling Applications

Digital Twins & Holographic Comms: Real-time digital mirroring and immersive communication.
Internet of Industrial Things (IIoT): Smart buildings, logistics, and supply chains.
Extended Reality (XR): Highly immersive virtual and augmented experiences.

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Thank You

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